

WASP-50b

R-0732

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_231221 · created 2026-07-28T01:03:23+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460299.83 +/- 0.1 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.05
Transit depth (Rp/Rs)^2	3.06 +/- 1.74 [%]
Orbital Inclination (inc)	87.31 +/- 2.93
Airmass coefficient 1 (a1)	0.8982 +/- 0.0088
Airmass coefficient 2 (a2)	0.069 +/- 0.022
Scatter in the residuals of the lightcurve fit is	0.98 %
Best Comparison Star	#2 - [180, 359]
Optimal Aperture	4.51
Optimal Annulus	10.44
Transit Duration (day)	0.022 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.05 vs 0.1404 (deviation 0.69σ)
Duration fit vs published	0.022 d vs 0.0754 d (deviation 1.84σ)
Mid-time O–C	168.6 min
Depth SNR	3.5
Residual scatter	0.98 %

Light curve

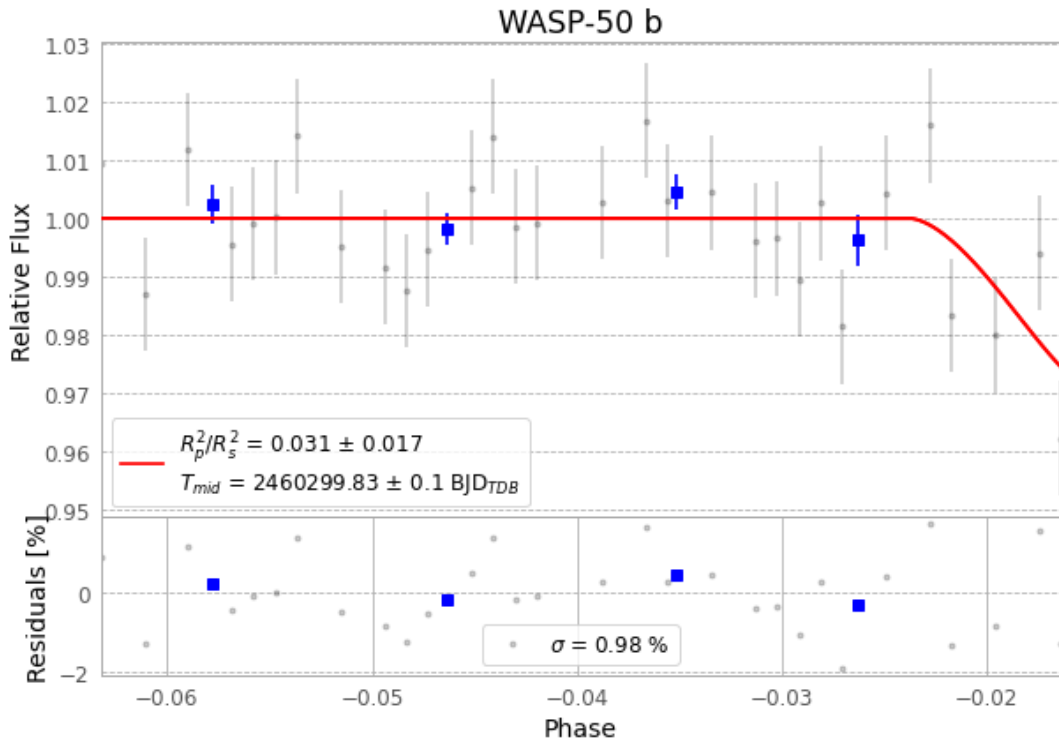


Figure 1 — FinalLightCurve_WASP-50 b_2023-12-20.png (SHA-256 3e21a0ec9526938a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [307, 389], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1d2a534eb82dbc70...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

31 FITS frames (20 MB), WASP-50231221044516.FITS → WASP-50231221065715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-231221.log (3c4295c22ac6...), FinalLightCurve_WASP-50 b_2023-12-20.png (3e21a0ec9526...), FinalLightCurve_WASP-50 b_2023-12-20.csv (1eb774ff472f...)

Compute resources

Wall time 70.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 01:03Z.